

EX: 3 DETERMINATION of VOLTAGE IN CIRCUIT
USING NODAL ANALYSIS

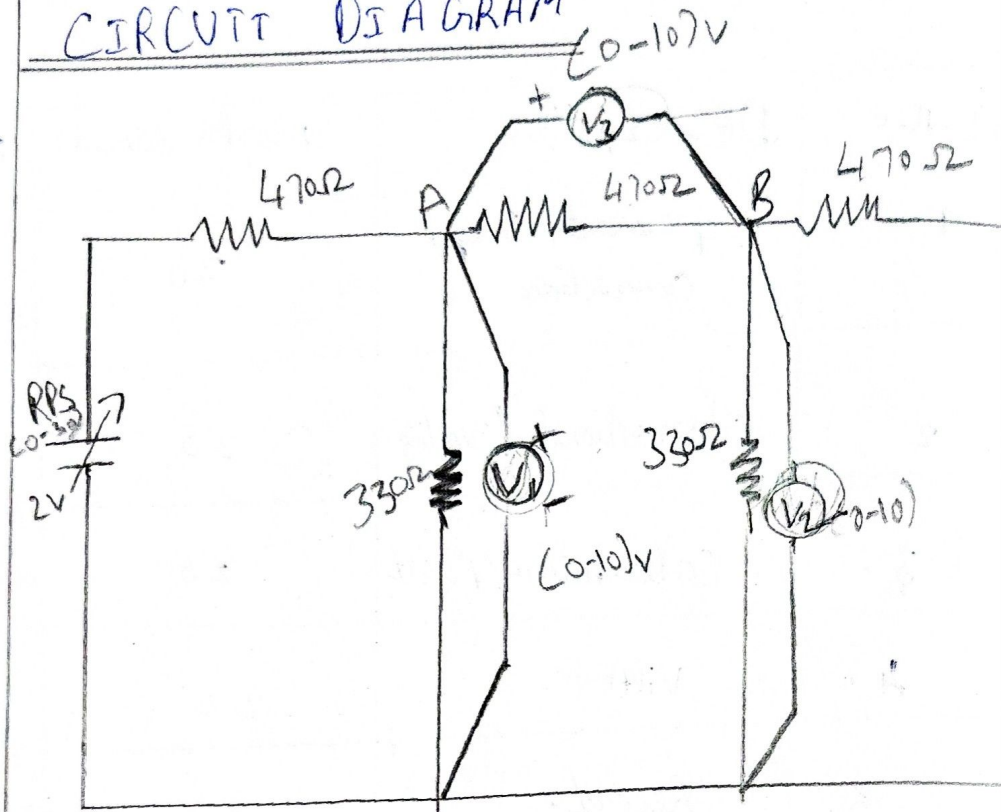
AIM:-

To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit

Apparatus Required :-

S.No	Apparatus	Specification	Quantity
1)	Regulated power supply (RPS)	(0-30)V	1
2)	Multimeter	--	1
3)	Resistors	470 Ω , 330 Ω	3, 2
4)	Bread board	--	1

CIRCUIT DIAGRAM



Theory :-

In electric circuits analysis, node-voltage analysis, or the branch current method is a method of determining the voltage (Potential difference) between "nodes" (Points where elements or branches connect) in an electrical circuit in terms of the branch currents.

Procedure :-

1. Give connections as per the circuit diagram.
2. Switch on the supply. Vary the RPs [Regulated Power Supply] and set a particular input voltage.
3. Note down the reading of ammeter and voltmeters and tabulate them.
4. Vary the RPs for different input voltages and note down the reading of all the meters.
5. Reduce the RPs to its minimum value and switch off the supply.
6. Using the tabulated values, verify Kirchhoff's laws practically and verify theoretically.

TABULAR COLUMN:-

	Parameters	Theoretical	Practical
	V_A	0.63	0.69
	V_B	0.452	0.49
	$V_A - V_B$	0.186	0.2

CALCULATIONS:-

Applying KCL at node A:-

$$i_1 = i_2 + i_3$$

$$i_2 = \frac{-2 - V_A}{470} ; i_3 = \frac{V_A - V_B}{470} ; V_2 = \frac{V_A}{330}$$

$$\frac{-2 - V_A}{470} = \frac{V_A - V_B}{470} + \frac{V_A}{330}$$

$$60 \cdot \frac{1}{4} \quad 47V_A - 33V_B = 66$$

$$113V_A - 33V_B = 60 \rightarrow (1)$$

Applying KCL at node B

$$i_5 = i_4 + i_5$$

$$i_4 = \frac{V_B}{330} ; i_5 = \frac{V_B}{470}$$

$$\frac{V_B - V_B}{470} = \frac{V_B}{330} + \frac{V_B}{470}$$

$$\frac{V_B - V_B}{470} = \frac{V_B}{330} + \frac{V_B}{470}$$

$$\frac{V_B}{330} = \frac{V_A - 2V_B}{470}$$

$$33V_A - 47V_B = 0$$

$$33 - 43V_B = 0$$

Eq ① & ②

$$\begin{bmatrix} 113 & -33 \\ 33 & -113 \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} 66 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 113 & -33 \\ 33 & -113 \end{vmatrix} = -12769 + 1089 = -11680$$

$$\Delta_1 = \begin{vmatrix} 66 & -33 \\ 0 & -113 \end{vmatrix} = -7458$$

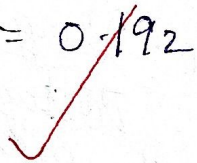
$$\Delta_2 = \begin{vmatrix} 113 & 66 \\ 33 & 0 \end{vmatrix} = -2178$$

$$V_A = \frac{\Delta_1}{\Delta} = \frac{-7458}{-11680} = 0.638V$$

$$V_B = \frac{\Delta_2}{\Delta} = \frac{-2178}{11680} = 0.186V$$

$$V_A - V_B = 0.638 - 0.486$$

$$= 0.192$$



SL.No	Description	Mark allotted	mark obtained
1.	EXPerimental Setup / connection	20 marks	18
2.	observation / Reading	20 marks	18
3.	Calculations / Result	20 marks	18
4.	VIVA	20 marks	18
5.	Record	20 marks	18
			<u>90</u>

Result :-

Thus, the nodal analysis is verified practically and theoretically. The resultant voltage for 2 V supply are

- The Voltage V_1 is 0.69V
- The Voltage V_2 is 0.49V
- The voltage V_3 is 0.2 V